# Muhammad Yousuf Rehan

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## SUMMARY

Computer Science graduate (FAST-NUCES, June 2026) with hands-on experience in **full-stack development**, **machine learning** and **test automation**. Built production multi-tenant web applications (**Next.js, React, TypeScript, Node.js, PostgreSQL**), integrated **Python/PyTorch** ML services into backends, and validated quality with **Playwright, Selenium and Postman**. Skilled in **Docker, CI/CD, Microsoft Azure** and Agile team environments.

## EDUCATION

**National University of Computer and Emerging Sciences (FAST-NUCES)** | Karachi, Pakistan — Graduated June 2026
*Bachelor of Science in Computer Science (BS CS)*
**Coursework:** Software Testing, Cloud Computing (Azure), Computer Vision, Recommender Systems, Database Management Systems, Information Security, Operating Systems, Parallel Programming (OpenMP), Data Structures and Algorithms, OOP, Theory of Automata

**BVS Parsi High School** | Karachi, Pakistan | Matriculation, Computer Science — May 2020

## EXPERIENCE

**Swenta Globals** | *Junior Full Stack Developer (Full-time)* — February 2026 – May 2026

- Developed features for **Apricart OneFlowe**, a multi-tenant, multi-branch inventory, order and budget platform, using **Next.js (App Router), React, TypeScript, Node.js, PostgreSQL, Drizzle ORM and Supabase**; designed schemas, migrations and role-specific dashboards (Tailwind CSS, shadcn/Radix UI, Recharts).
- Implemented a **4-tier RBAC** model (Super Admin, Head Office, Branch Admin, Order Portal) and **tenant isolation with PostgreSQL Row-Level Security** through a custom withTenant wrapper binding organization and branch context to queries.
- Built the order workflow with an **approval-token (TID)** mechanism validating state transitions (e.g., PENDING to FULFILLED) and the "Smart Sync" analytics layer aggregating fulfillments, rejections and refunds for multi-branch executive reporting.
- Integrated a **Python/PyTorch restocking-prediction API** with the Next.js backend to serve per-branch stock forecasts from time-series sales data.
- Tested access control (RBAC, RLS), order lifecycle edge cases, analytics accuracy and the Next.js–Python integration; automated bulk user generation and checking with **Selenium**; used Docker, Git/GitHub and Mermaid.js documentation.

**Basil Innovations Ltd.** (Remote), Karachi, Pakistan | *Junior Developer (Part-time/Contract)* — January 2023 – May 2024

- Developed and optimized **RESTful APIs (Python, Node.js)** for fintech-oriented internal products in production, improving response time, data validation and reliability.
- Designed relational **MySQL** schemas for data consistency, normalization and efficient queries; containerized services with **Docker** and helped maintain **CI/CD** pipelines.
- Tested APIs and backend workflows with **Postman**, catching edge cases and reducing runtime errors; used Git/GitHub branching and pull requests; took part in sprint planning, code reviews and documentation.

## PROJECTS

- **Final Year Project – Self-Learning Weather Prediction Agent** | *Python, Reinforcement Learning, Random Forest, LightGBM*: Built a self-learning weather prediction system; compared Random Forest and LightGBM models and designed preprocessing pipelines for real-world data.
- **AI-Powered Book Recommendation System** | *Collaborative Filtering*: Generated personalized book recommendations from the GoodReads dataset.
- **SmoothGrad – CNN Interpretability** | *Computer Vision*: Implemented SmoothGrad to reduce noise in CNN saliency maps by adding noise to inputs and averaging the results.
- **Student Learning & Course Scheduling Platform** | *Microsoft Azure*: Built a full-stack platform with authentication, course enrollment and scheduling; deployed backend services and databases on Azure.
- **SY Fitness Club – Gym Management System** | *React, Node.js, Express, MySQL*: Built role-based access, real-time registration and package management; produced SRS, ERD, use case, class and sequence diagrams.
- **Software Testing Course Project** | *Playwright*: Applied automated and manual testing to validate application behavior.
- **Custom C-Like Compiler** | *Compiler Construction*: Implemented lexical, syntax and semantic analysis with custom data types and validation rules.

## TECHNICAL SKILLS

**Languages:** JavaScript, TypeScript, Python, Java, C, C++, SQL
**Web & Backend:** React, Next.js (App Router), Node.js, Express.js, RESTful APIs, Tailwind CSS, shadcn/ui, Recharts, HTML5, CSS3
**Databases:** PostgreSQL (Row-Level Security), MySQL, MongoDB, Supabase, Drizzle ORM
**Testing & QA:** Playwright, Selenium, Postman, Manual and Automated Testing, Test Case Design, Edge-Case Analysis
**Machine Learning & AI:** PyTorch, Random Forest, LightGBM, Supervised & Unsupervised Learning, Reinforcement Learning (Q-Learning), Neural Networks, CNNs, Computer Vision, Recommender Systems, Time-Series Forecasting, Model Interpretability, Agentic AI
**Cloud, DevOps & Tools:** Microsoft Azure, Docker, CI/CD, Git, GitHub, Mermaid.js, VS Code
**Concepts:** RBAC, Multi-Tenant Design, Authentication, Information Security, Data Structures and Algorithms, OOP, Operating Systems, DBMS, Parallel Programming (OpenMP), Agile / Sprint Planning

## ACTIVITIES

Participated in university coding competitions, coder cups and hackathons (certificates of participation and achievement); continuously upskill through online courses and hands-on projects.